

# End-to-End Predictive Solution to Reduce Late Deliveries and Improve Customer Satisfaction at Olist

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# EXECUTIVE SUMMARY

## BUSINESS PROBLEM

Olist needs to identify orders at risk of late delivery before the delivery process begins.
**Late delivery is an important customer experience issue.** The average review score drops from **4.21 for on-time orders to 2.26 for late orders,** and 53.8% of late orders receive 1 star.

## KEY FINDINGS

Late delivery is associated with several factors:

- Sellers missing their dispatch deadline
- Shorter promised days
- Bigger volume
- Seasonal events

## MODEL

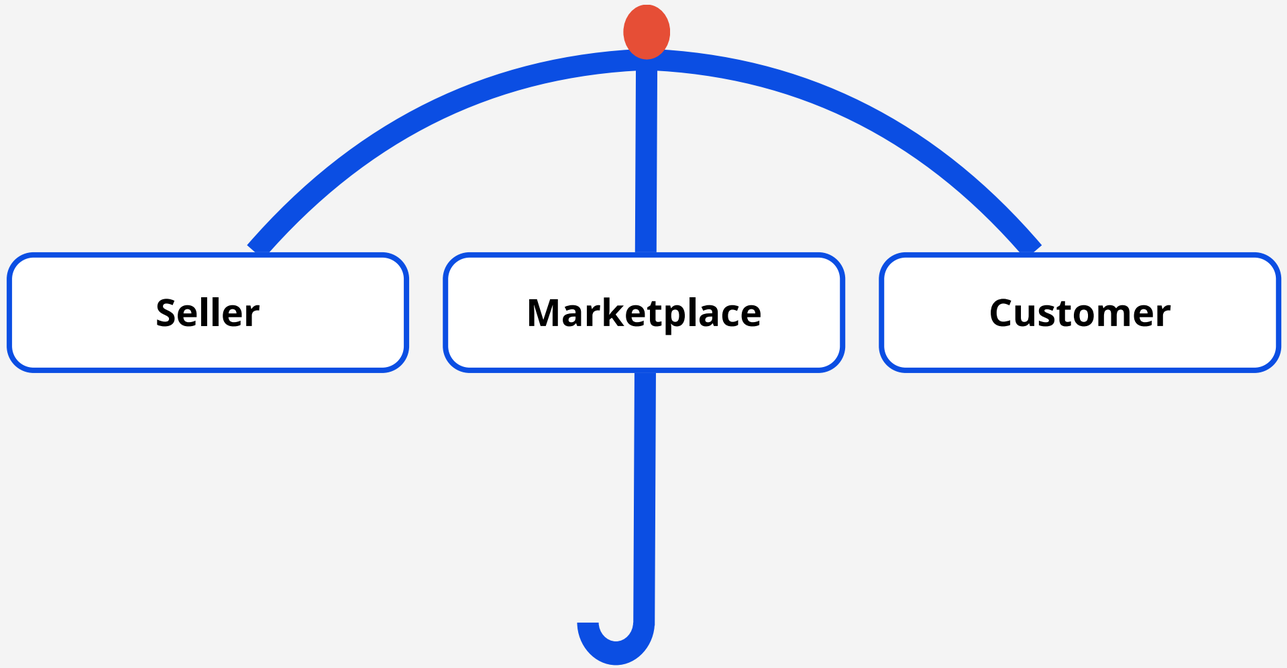
The model was developed using 95,070 orders, with features available at order approval. **Tuned LightGBM** selected as final model (**PR-AUC 0.24, ROC-AUC 0.76**) optimized to prioritize capturing potential late orders.

## BUSINESS IMPACT

The proposed targeted intervention scenario could:

- **Reduce late delivery** from 5.4% → 1.18%
- **Improve average review score** by 32,26%
- **Generate net value** estimated R$15,793.24

# COMPANY BACKGROUND

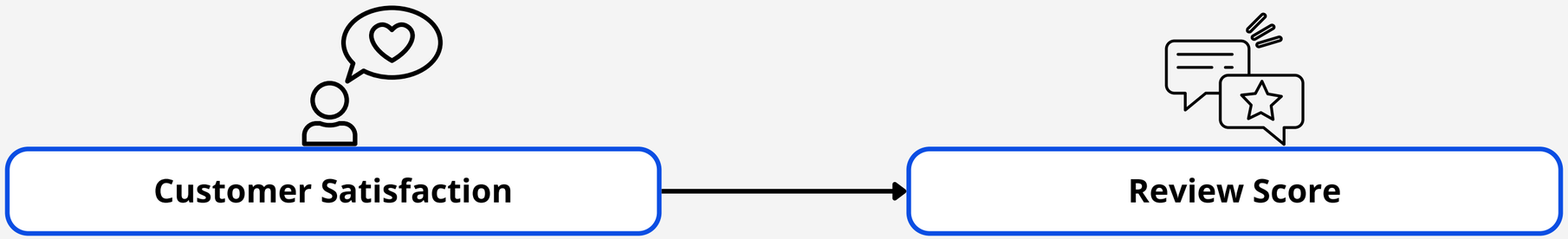



Olist is a **Brazilian e-commerce ecosystem** that provides integrated solutions to support online sales, logistics, and customer experience.

Olist's core business:

- E-commerce ecosystem
- Products: Olist Store, Olist Shops, logistics services (Envios da Olist / Olist Pax), financial solutions (Olist Credit, Olist Pay), and enterprise resource planning (ERP) integrations

# Olist's Success Meter: Customer Satisfaction

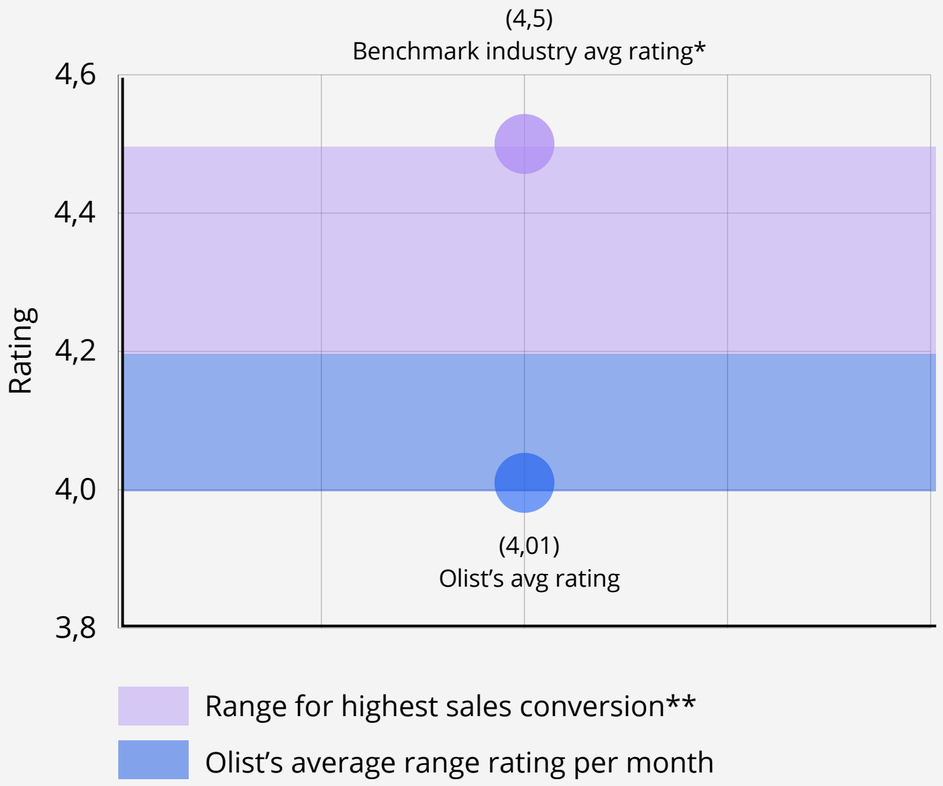


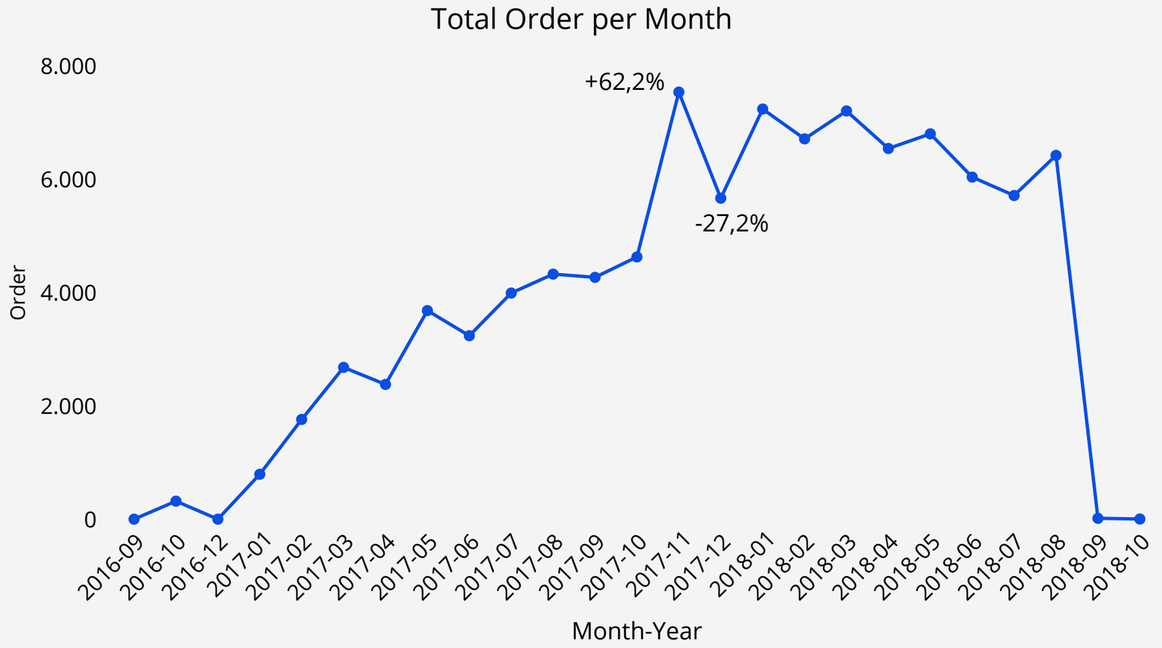

**Why Customer Satisfaction matters:**
Better customer experience can increase customer trust.
Higher trust can support more purchases and revenue.

**Olist's AVG review score:**

**4.01**

# Olist’s 4.01 Rating Is Below The Ideal Range and May Limit Olist’s Growth



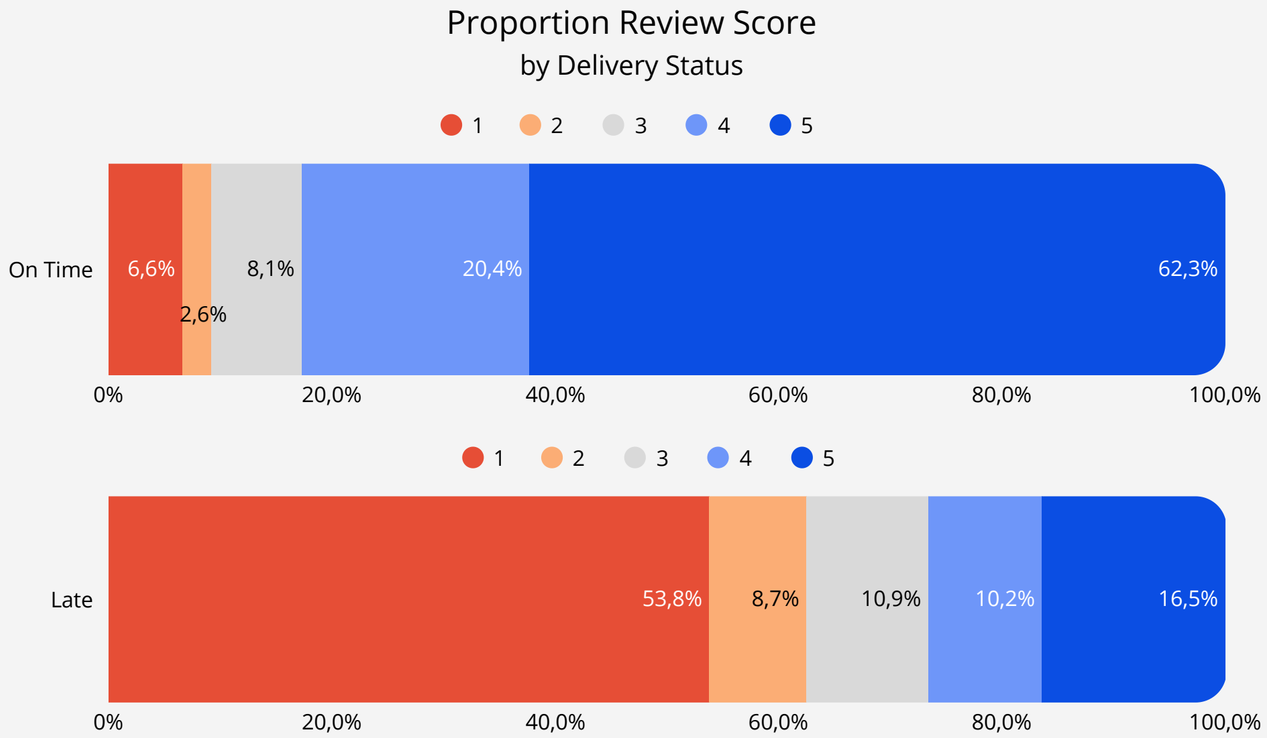

Olist’s order volume increases, resulting 20x growth but satisfaction is not improving. Also, the order volume incerase is not followed by stable growth.

* PowerReviews (2022)
** Northwestern University, Spiegel Research Center

# Late Delivery Is a Key Driver of Lower Review Score

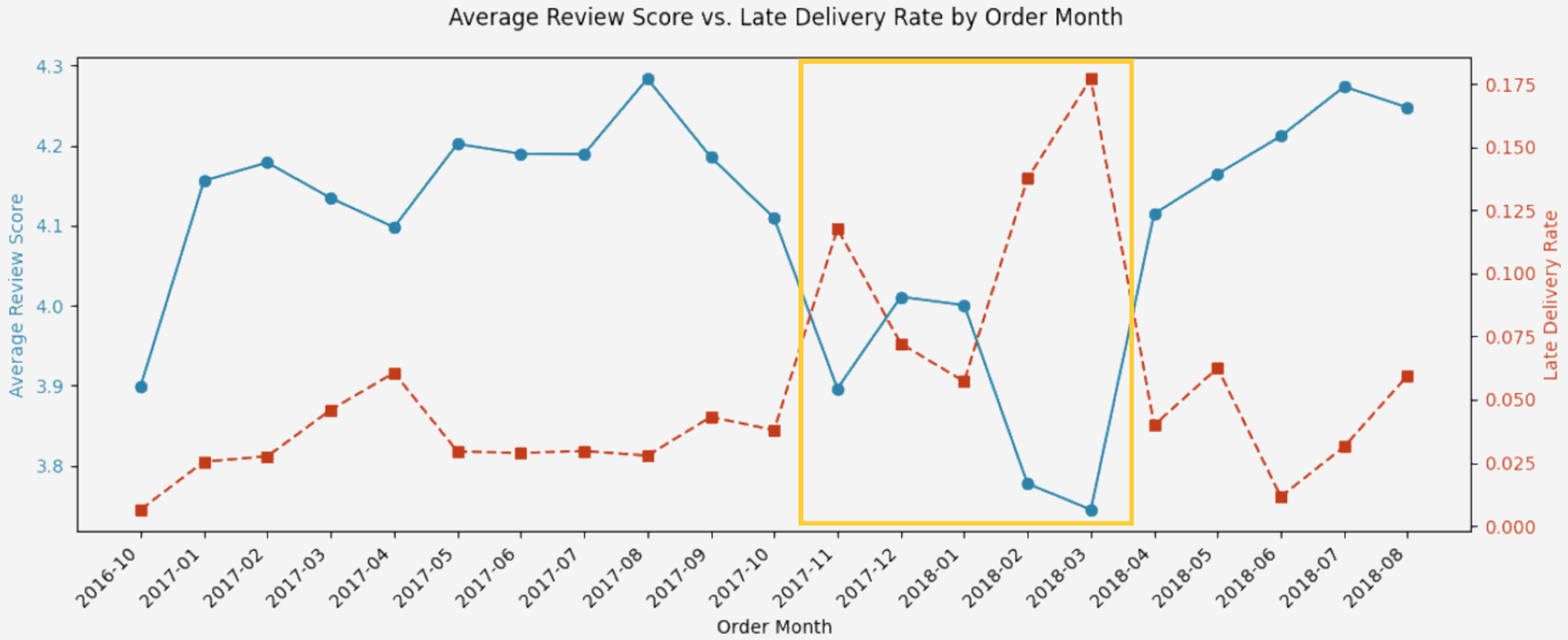
Review Score Comparison: Late vs On-Time

| ON-TIME | LATE |
|---|---|
| 4.21 ⭐ | 2.26 ⭐ |




Late deliveries are strongly associated with poor customer reviews, so Olist should use the model to identify high-risk orders early and prioritize seller or logistics interventions before delays impact the customer experience.

# Late Delivery and Lower Review Score Pattern Remains Consistent




**Orders with higher late-delivery rates tend to receive lower review scores**, and this relationship remains consistent throughout the observed period, making early identification and intervention on high-risk orders important for improving customer satisfaction.

# PROBLEM STATEMENT

Olist faces **low customer review scores** due to **late deliveries**

# BUSINESS STAKEHOLDER

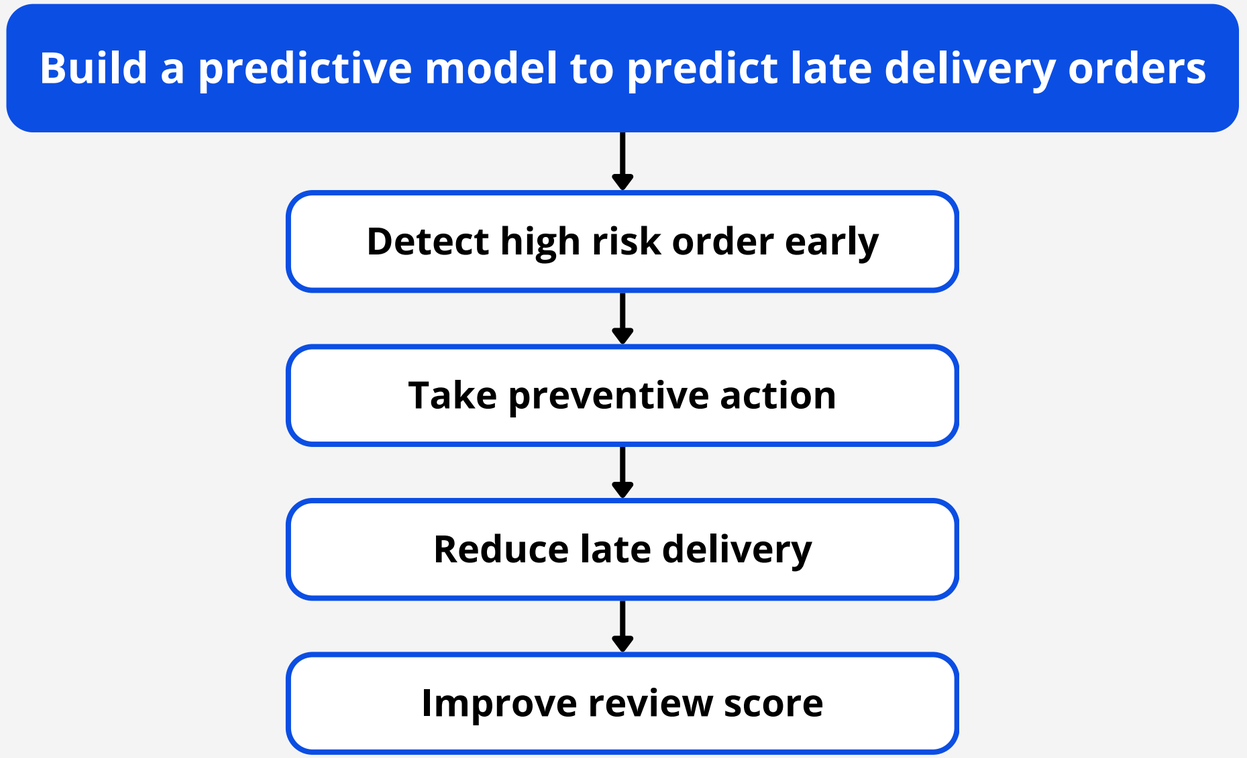


**Key stakeholder:**
Olist's Operation & Logistic Team

# BUSINESS QUESTIONS

1. What factors are associated with customer's late delivery?
2. How can a machine learning model predict which orders are at risk of late delivery early enough for Olist to act?

# GOALS


Build a predictive model to predict late delivery orders
Detect high risk order early
Take preventive action
Reduce late delivery
Improve review score

# DATA UNDERSTANDING

## DATASET SOURCE

Brazilian E-Commerce Dataset by Olist

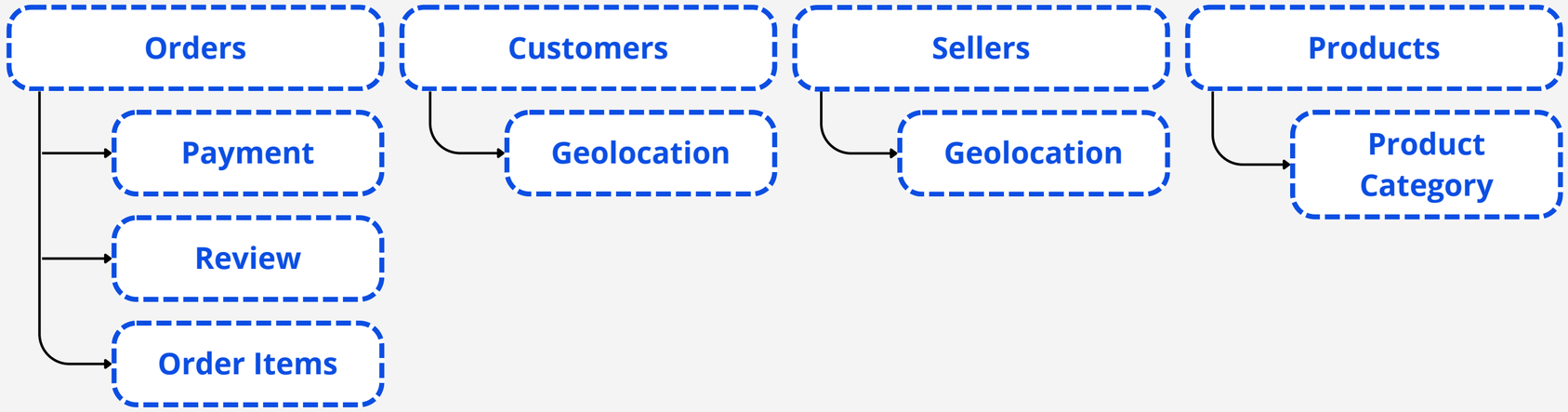
ADDED

Brazilian Zip Codes Coordinates

## DATASET LIMITATION

SEP 2016 - OCT 2018

## DATASET INFORMATION



# DATA OVERVIEW

**4,01** AVG REVIEW SCORE

**3095** TOTAL SELLERS

**94737** TOTAL CUSTOMERS

**98017** TOTAL ORDERS

# DATA ATTRIBUTES

**ORDERS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| order_id | string | Unique identifier of the order (an order can have multiple items) |
| customer_id | string | Key to the customer who placed the order. |
| order_status | string | Order lifecycle status (delivered, shipped, canceled, etc). |
| order_purchase_timestamp | datetime (string) | Timestamp when the order was placed. |
| order_approved_at | datetime (string) | Timestamp when payment was approved. |
| order_delivered_carrier_date | datetime (string) | Timestamp when the order was handed to the logistics carrier. |
| order_delivered_customer_date | datetime (string) | Timestamp when the order was actually delivered to the customer. |
| order_estimated_delivery_date | datetime (string) | Delivery date estimate given to the customer at purchase time. |

# DATA ATTRIBUTES

**ORDER_ITEMS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| order_id | string | Unique identifier of the order. |
| order_item_id | integer | Sequential number identifying number of items within the order (1, 2, 3...). |
| product_id | string | Product unique identifier. |
| seller_id | string | Seller unique identifier. |
| shipping_limit_date | datetime (string) | Seller's deadline to hand the item to the logistic partner (carrier). |
| price | float | Item price. |
| freight_value | float | Freight/shipping cost for this item. (if an order has more than one item the freight value is splitted between items) |

# DATA ATTRIBUTES

**CUSTOMERS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| customer_id | string | key to the orders dataset. Each order has a unique customer_id. |
| customer_unique_id | string | The actual unique person (use this to track customers across multiple orders) |
| customer_zip_code_prefix | integer | First 5 digits of customer's zip code. |
| customer_city | string | Customer's city. |
| customer_state | string | Customer's state (Brazilian state code, e.g. SP, RJ). |

**SELLERS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| seller_id | string | Unique identifier of the seller. |
| seller_zip_code_prefix | integer | First 5 digits of seller's zip code. |
| seller_city | string | Seller's city. |
| seller_state | string | Seller's state. |

# DATA ATTRIBUTES

**PAYMENTS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| order_id | string | Unique identifier of the order. |
| payment_sequential | integer | Sequence number if a customer used more than one payment method for the same order. |
| payment_type | string | Payment method (credit_card, boleto, voucher, debit_card). |
| payment_installments | integer | Number of installments chosen. |
| payment_value | float | Amount paid via this payment row. |

# DATA ATTRIBUTES

**ORDER_REVIEWS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| review_id | string | Unique identifier of the review. |
| order_id | string | Unique identifier of the order. |
| review_score | integer | Customer's satisfaction rating, 1 (worst) to 5 (best). |
| review_comment_title | string | Title of the written review left by the customer (in portuguese). |
| review_comment_message | string | Comment message left by the customer (in portuguese). |
| review_creation_date | datetime (string) | Date the satisfaction survey was sent to the customer. |
| review_answer_timestamp | datetime (string) | Timestamp when the customer submitted the review. |

# DATA ATTRIBUTES

**PRODUCTS**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| product_id | string | Unique identifier of the product. |
| product_category_name | string | Product category name (in portuguese). |
| product_name_lenght | float | Number of characters in the product name. |
| product_description_lenght | float | Number of characters in the product description. |
| product_photos_qty | float | Number of photos published for the product. |
| product_weight_g | float | Product weight in grams. |
| product_length_cm | float | Product length in cm. |
| product_height_cm | float | Product height in cm. |
| product_width_cm | float | Product width in cm. |

# DATA ATTRIBUTES

**PRODUCT_CATEGORY_NAME_TRANSLATION**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| product_category_name | string | Category name in Portuguese. |
| product_category_name_english | string | Same category translated to English. |

**GEOLOCATION**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| geolocation_zip_code_prefix | integer | First 5 digits of the zip code. |
| geolocation_lat | float | Latitude of the zip code location. |
| geolocation_lng | float | Longitude of the zip code location. |
| geolocation_city | string | City name for the zip code. |
| geolocation_state | string | State abbreviation for the zip code. |

# DATA PREPARATION

**LOAD DATASET**

**MERGE DATASET** Datashape (before): 113425, 47

**DATA CLEANING**

### DUPLICATE DATA

- No exact duplicate rows were found
- 13984 repeated order_ids were validated as legitimate multi-item orders and retained.

### DATA CONCISTENCY

Fixed spelling, check data consistency and standardized name across several columns:

- **Product category**
- **Payment type**
- **State**
- **City**
- **City canonicalization**
- **Mapping**:

### MISSING VALUE

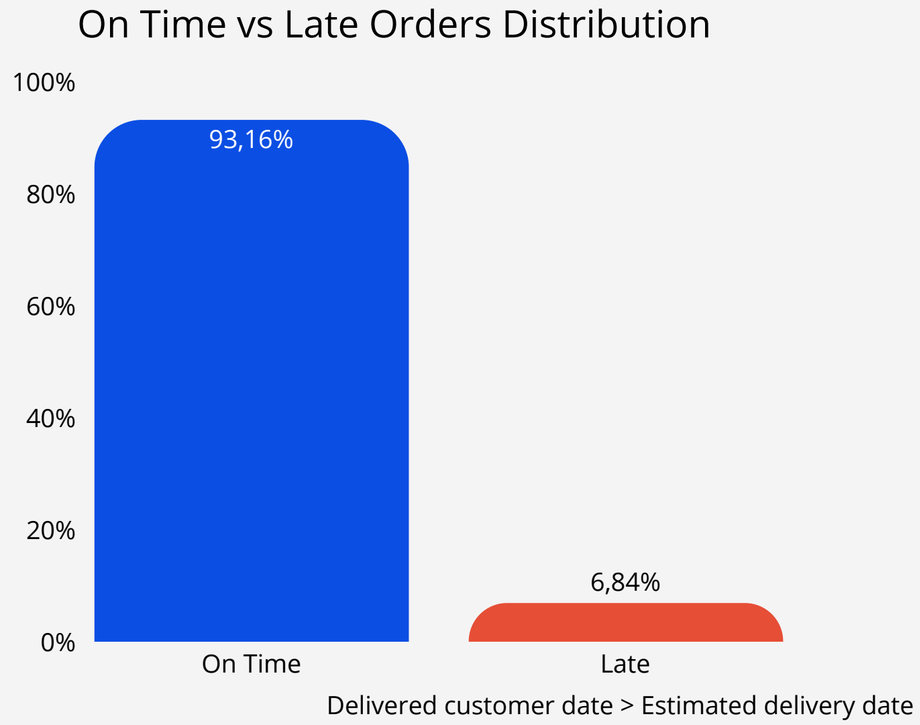
- Filled missing value for locations, product category name translations
- Drop missing value within payment details, date information
- Retained missing value in seller information

### LOGICAL FALLACY

Validated the chronological order of fulfillment timestamps and removed 1,400 orders with logical inconsistencies.

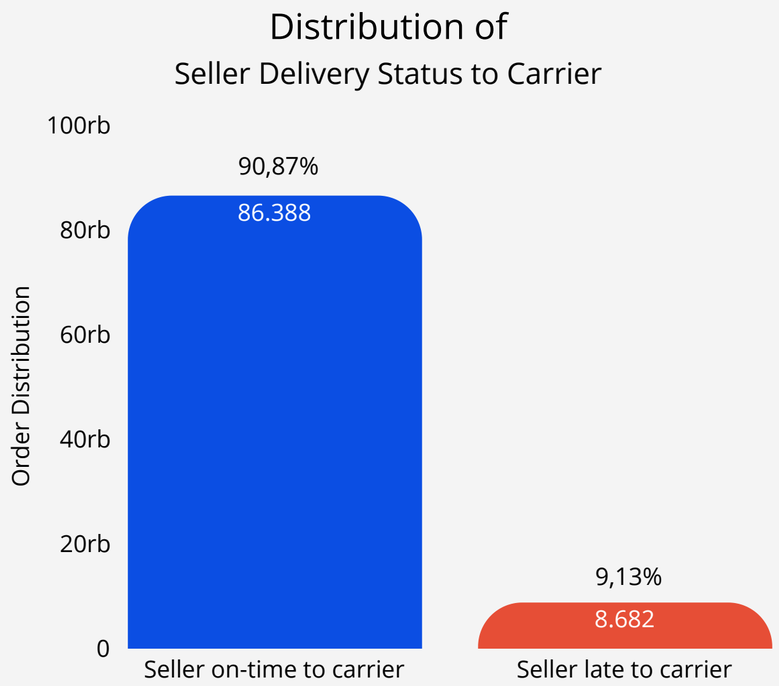
**EDA** Datashape (after): 111777, 39

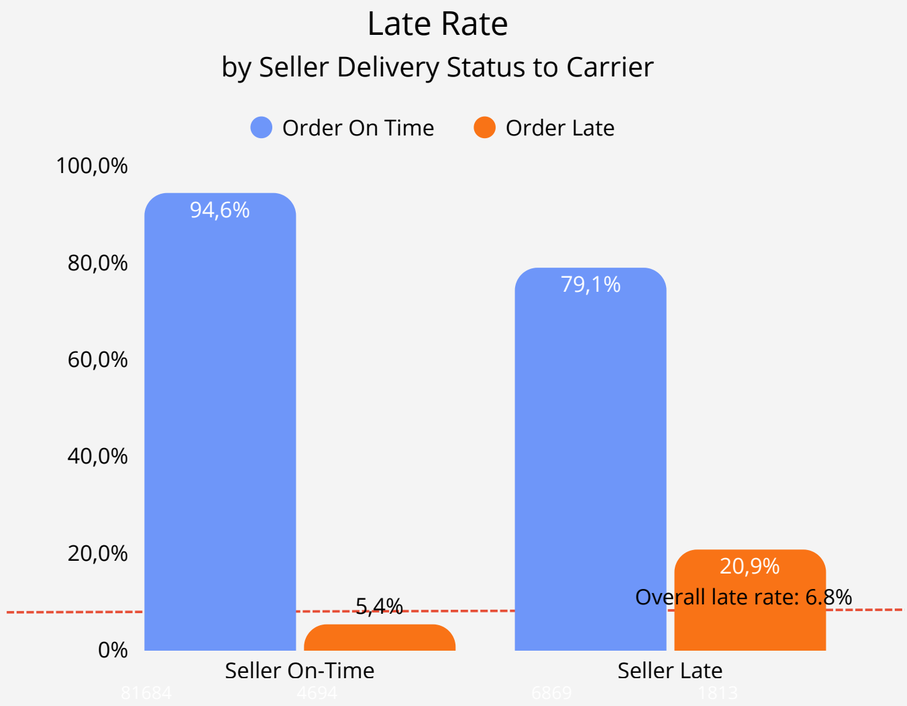
# Most Orders Are Delivered, with 6.84% Experiencing Late Delivery




With a **6.84% late-delivery rate** affecting **more than 6,300 customers,** Olist should prioritize early identification of high-risk orders and take targeted action before delays impact the customer experience.

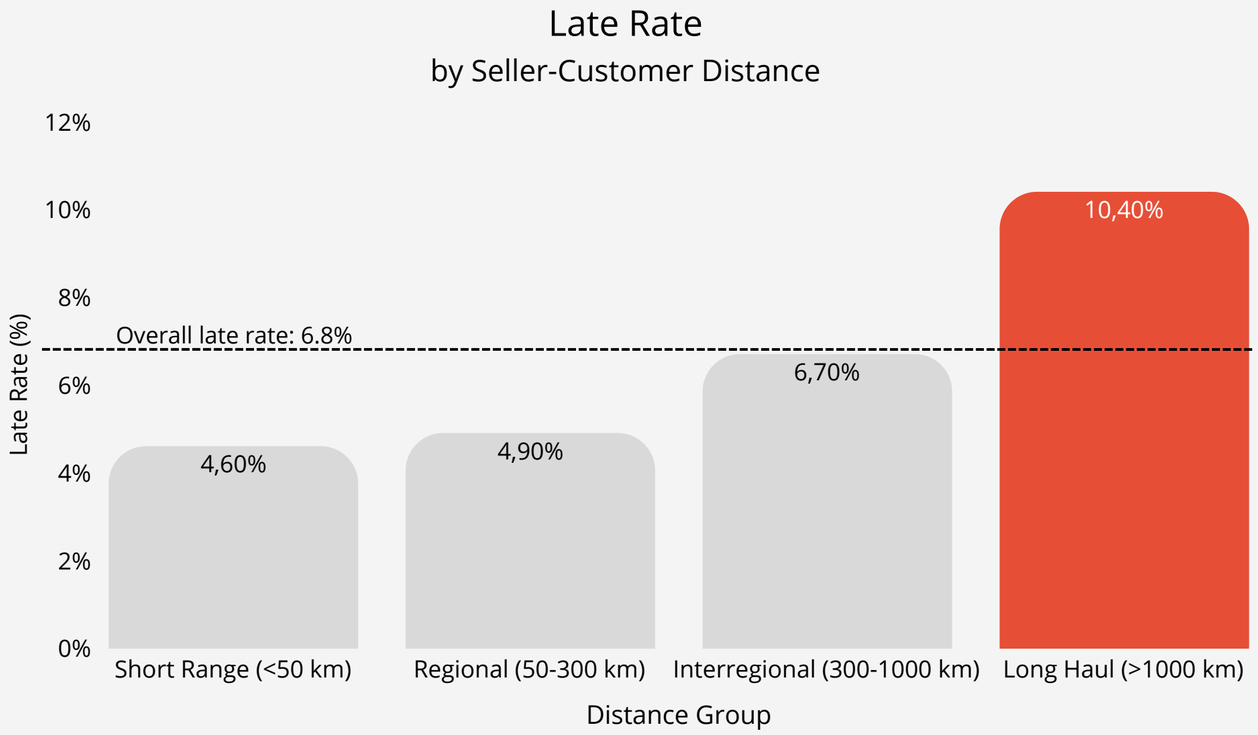
# Late Delivery Risk Increases When Sellers Delay Dispatching Orders

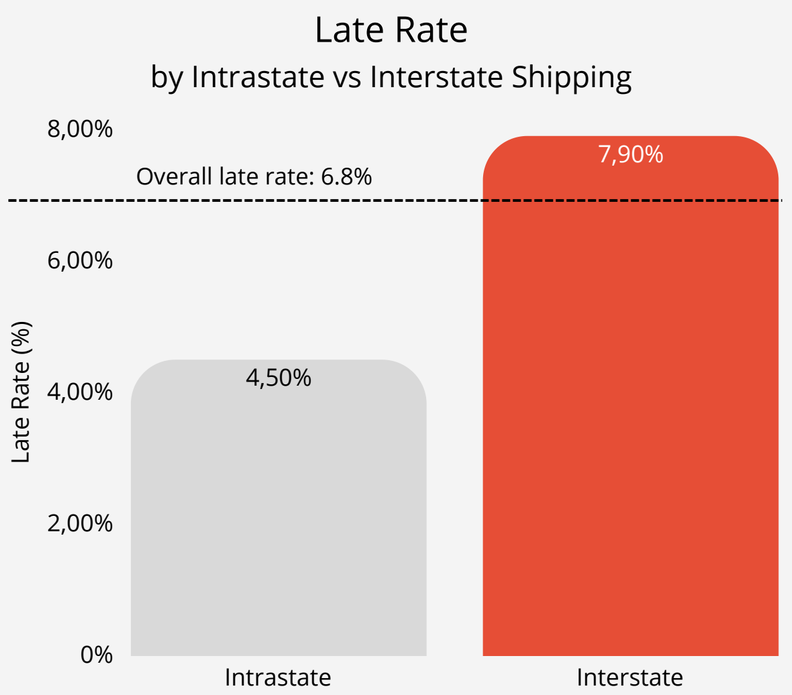



Most sellers (90.87%) **dispatch orders on time**, but **missing the dispatch deadline increases late-delivery risk** from 5.4% to 20.9%. However, since **most late orders still come from sellers who dispatch on time**, seller delays should be monitored as an early warning signal while Olist also addresses other transit and logistics factors through targeted early intervention.

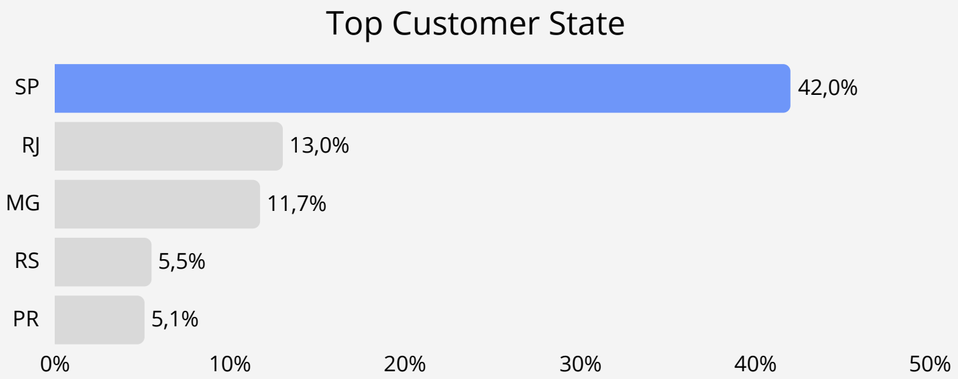
# The Farther the Order, the Higher the Late Rate

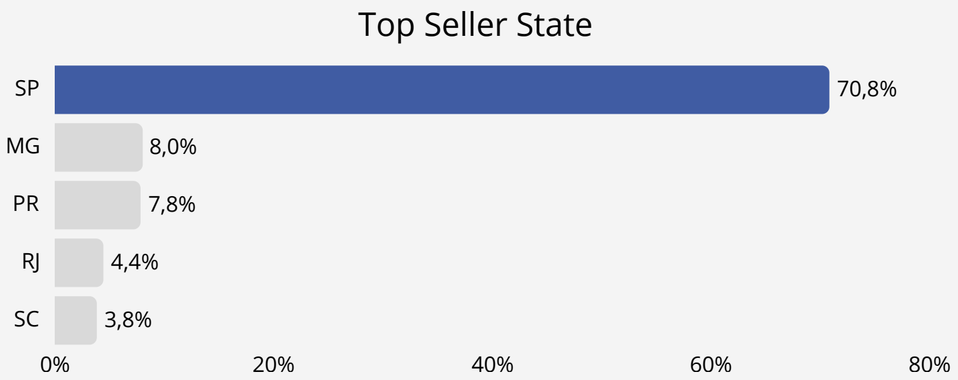


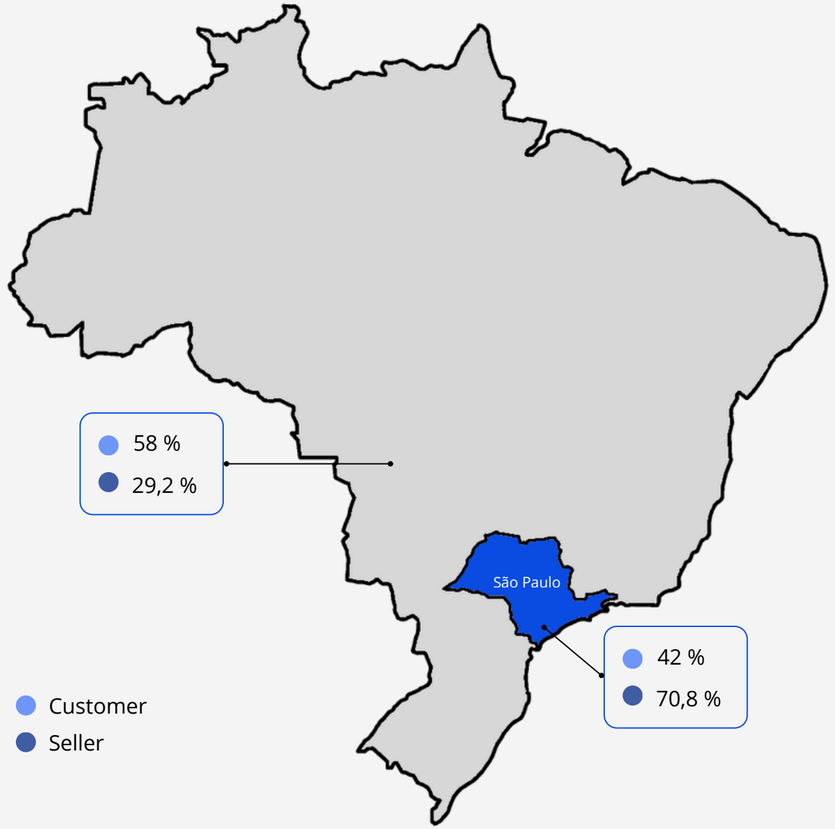




**Long-haul orders**, especially those traveling over 1,000 km **and across states**, **face higher late-delivery risk**, so Olist should prioritize these routes for early monitoring and consider faster shipping or logistics interventions for high-risk orders.

# Sellers and Customers Share Similar Geographic Concentration




**Seller and customer are concentated in the same major states**, with different level of concentration and sellers more concentrated in São Paulo, which can **create longer seller-to-customer distances** and increase delivery risk. To reduce late deliveries, the logistics team should prioritize long-distance routes for early monitoring and provide faster shipping options or route optimization for high-risk orders.

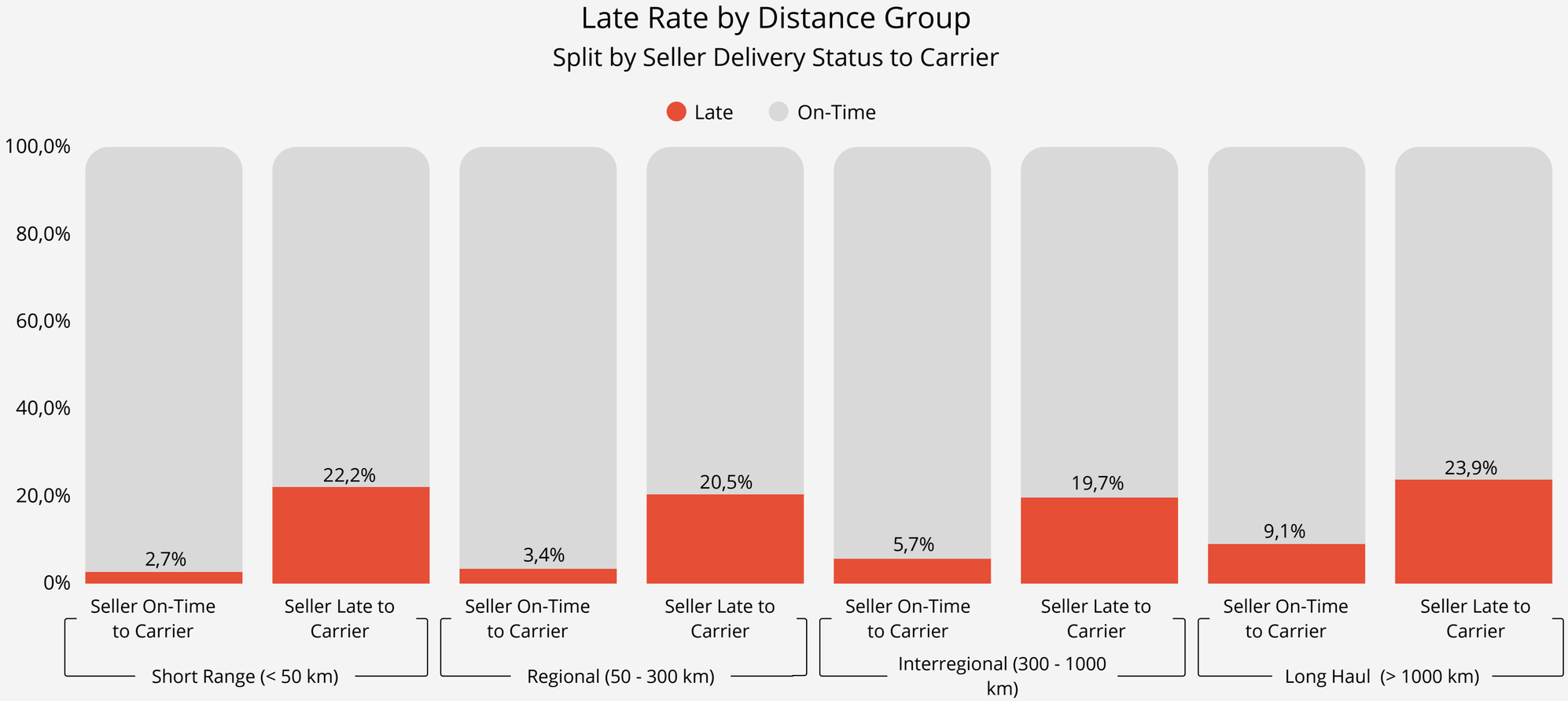
# Late Delivery Risk Is Higher on Longer and Cross-State Routes




| RISKY ROUTE | SAFEST ROUTE |
|---|---|
| SP > RJ | RS > SP |
| SP > BA | MG > MG |
| SP > MA | MG > SP |
| SP > AL | PR > SP |
| SP > CE | SP > SP |

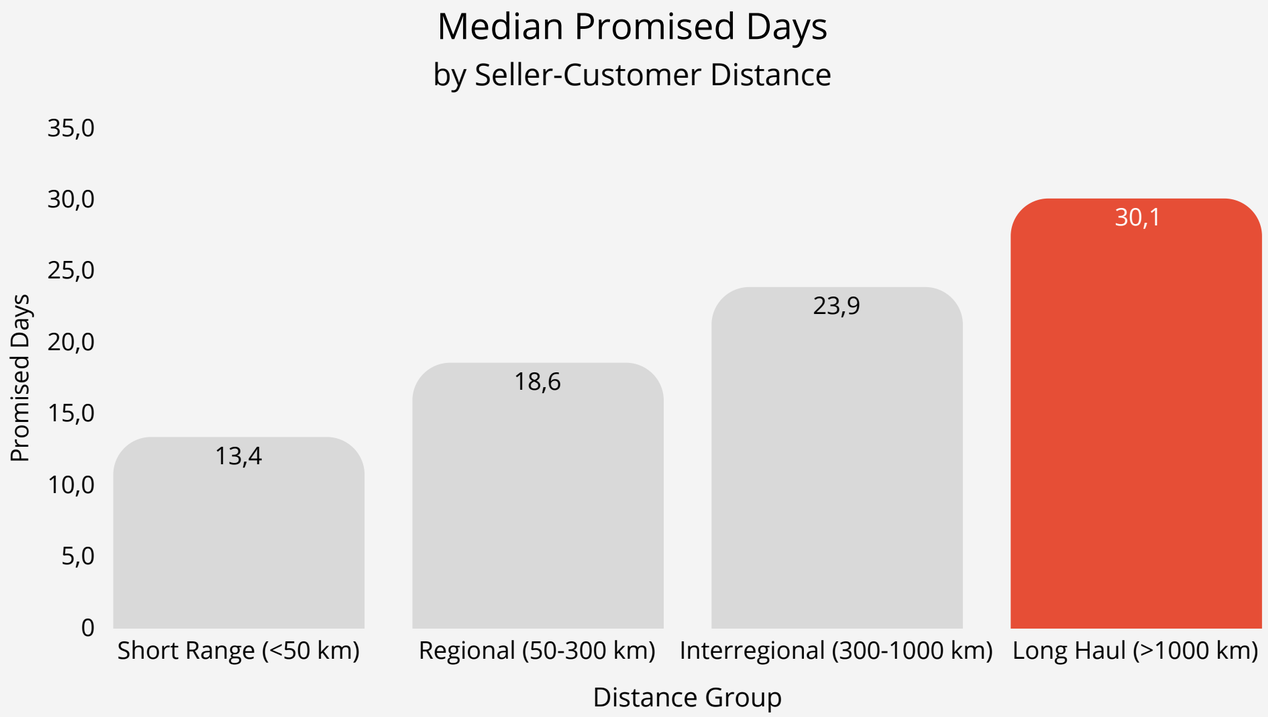
- High-risk routes are mainly shipments from São Paulo to other states
- Intrastate routes such as SP→SP and MG→MG have lower risk,
- This suggesting that distance, logistics access, and seller concentration in São Paulo may contribute to delivery delays
- Logistics team should prioritize high-risk routes for earlier monitoring and consider faster or more reliable shipping options to reduce late deliveries.

# Seller Dispatch Delays Impact All Delivery Distance



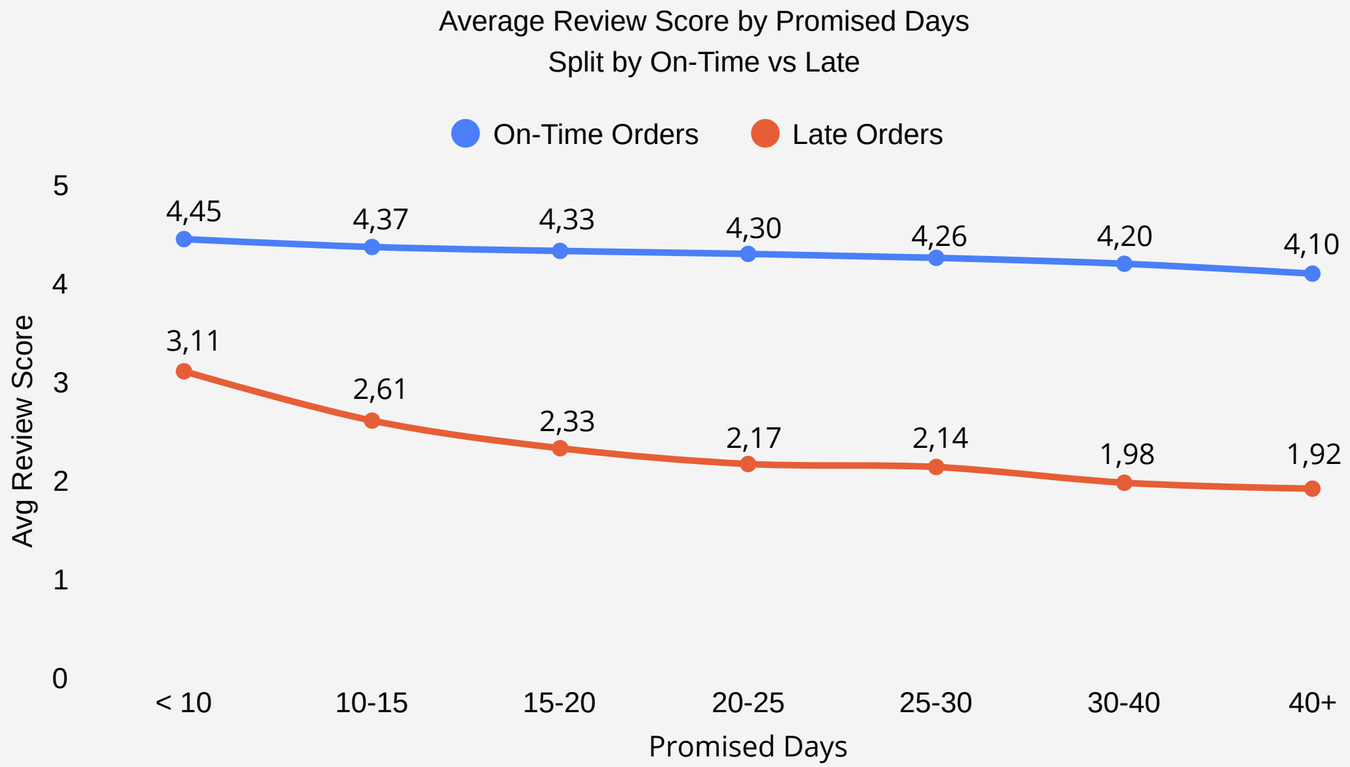

- Seller delays increase late-delivery risk across all distance groups.
- Late-delivery rates increase with distance, even when sellers dispatch on time.
- The highest risk occurs when seller lateness combines with long-distance routes
- Olist should prioritize these orders for early monitoring and intervention.

# Distance Is The Strongest Factor Shaping Promised Days



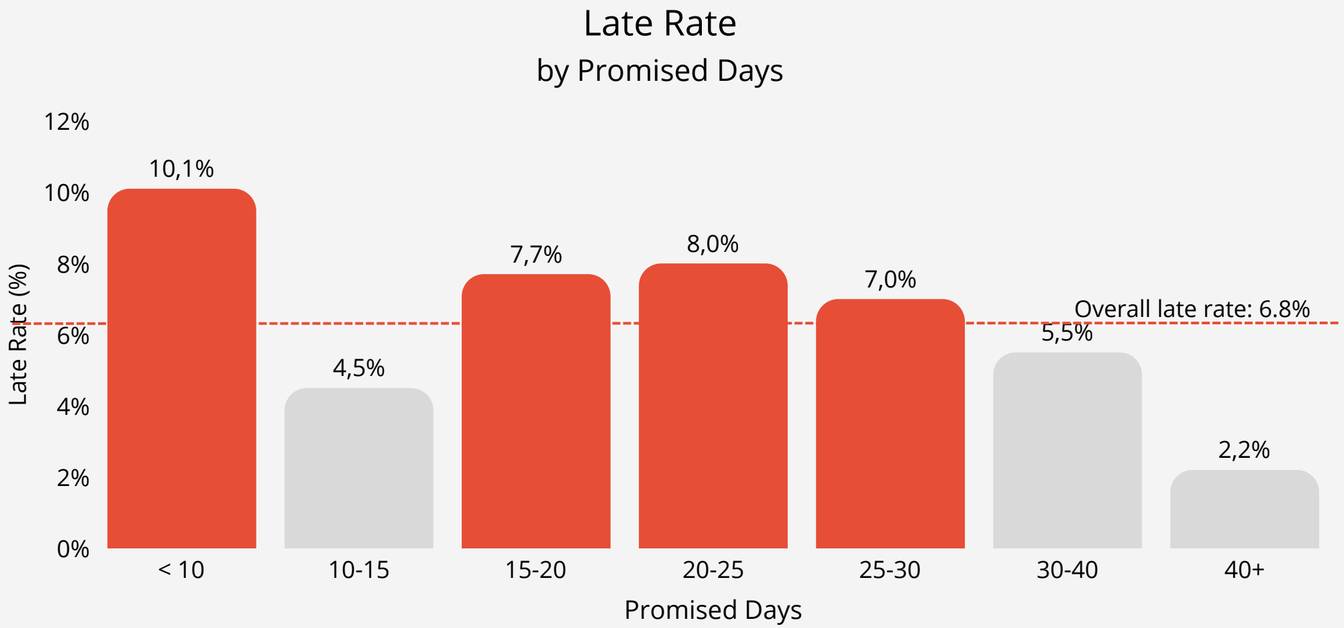

The median promised days show that **Olist adjusts its delivery estimates based on order distance**, with promised days increasing from short-range (<50 km) to long-haul deliveries (> 1000 km).

# Actual Delivery Performance Matters More Than Promised Days



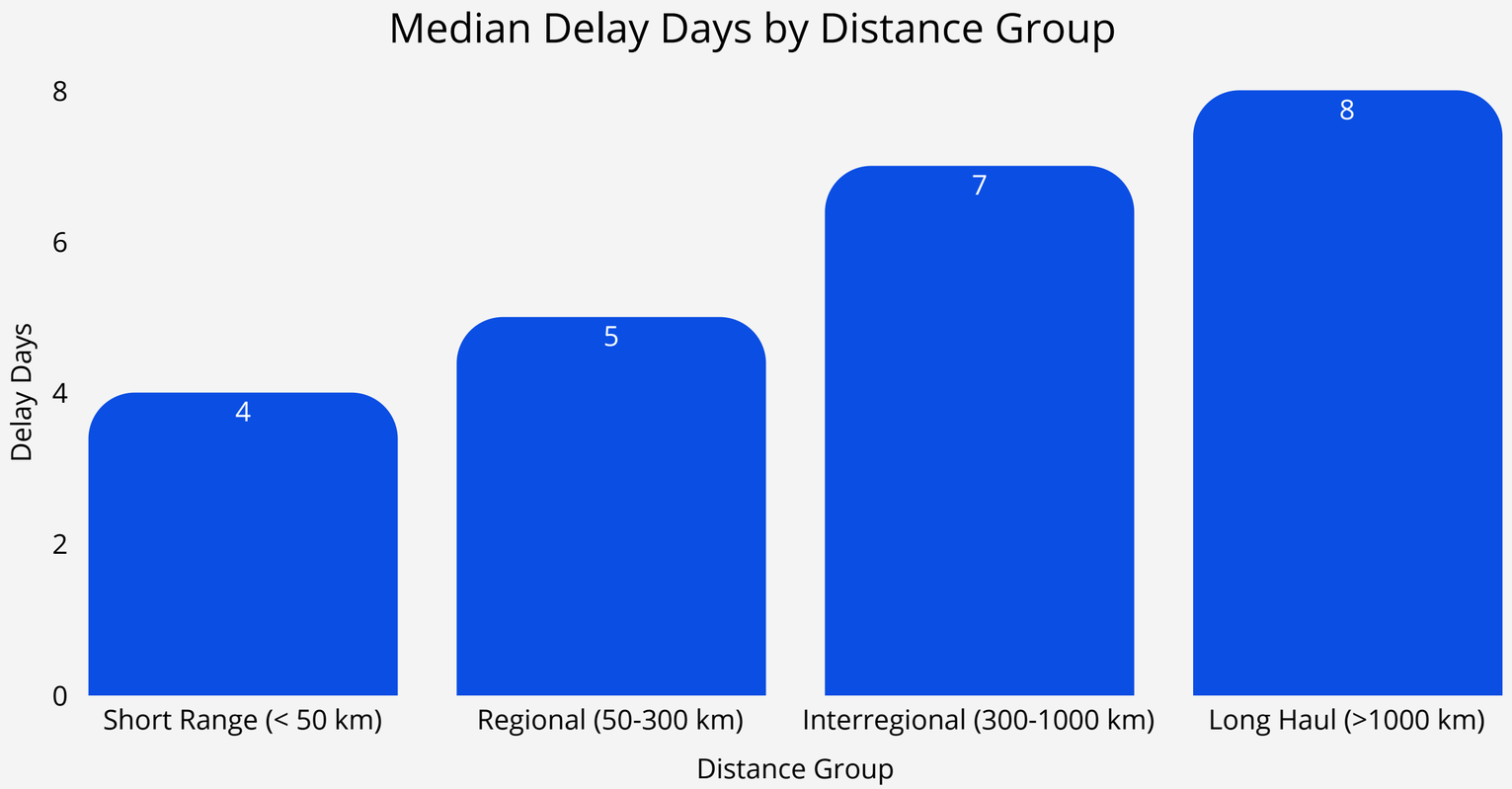

As **promised delivery times increase, review scores tend to decline**, but **late orders impacting the review score more than longer promised days** so Olist should focus on keeping actual delivery within the promised window and prioritize high-risk orders for early intervention to protect customer satisfaction.

# Shorter Promised Delivery Days Have Higher Late Delivery Risk



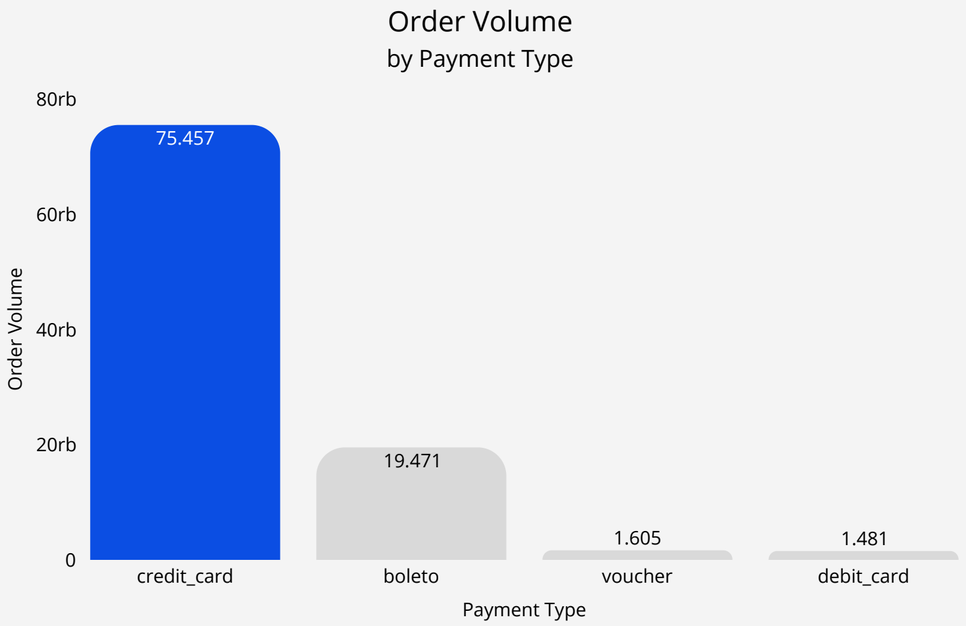

**Shorter promised delivery windows carry higher late-delivery risk** because they leave less buffer for unexpected delays. Tight delivery windows require more caution, not because faster delivery is inherently riskier, but because there is less tolerance for delays so Olist should **set realistic delivery estimates** and closely monitor orders with tight delivery windows.

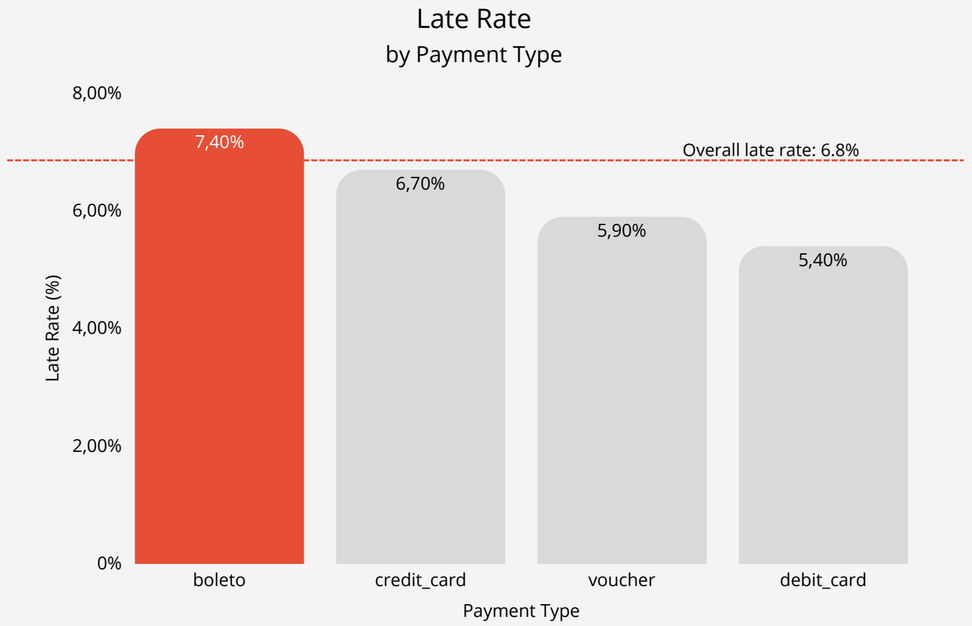
# Longer Routes Show Higher Median Delay Days




Longer-distance orders tend to experience longer delays, so **use distance-based median delay patterns** as a reference to **set more realistic delivery estimates** and customer expectations.

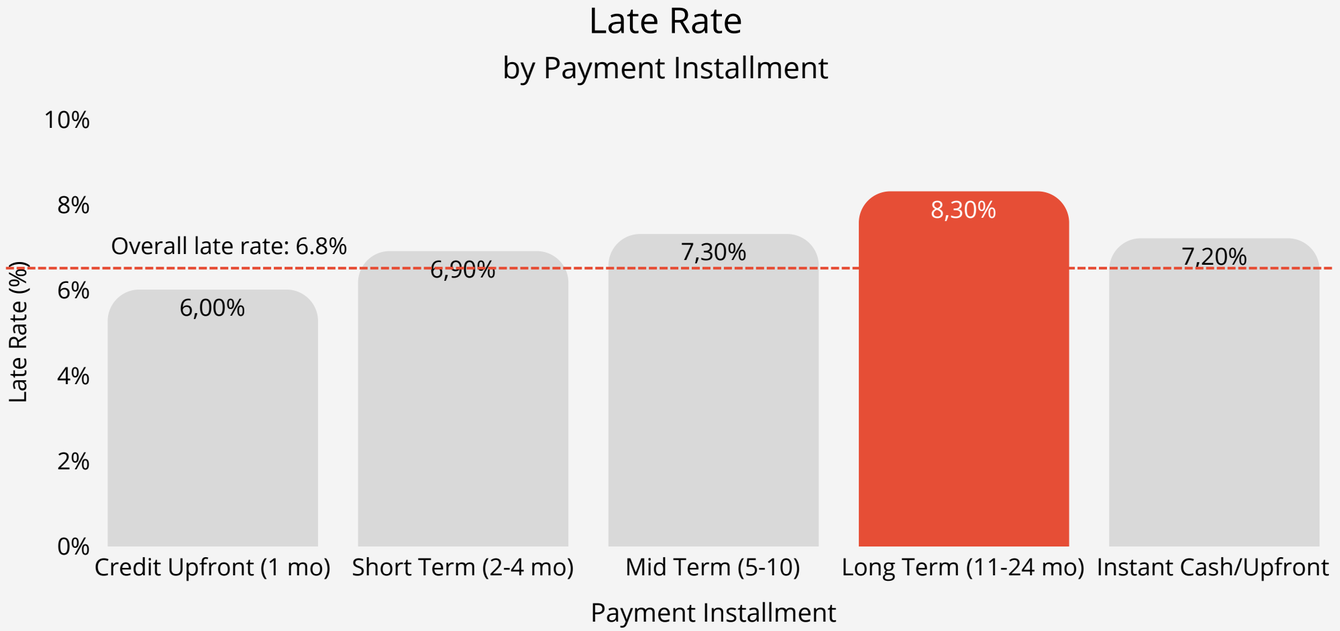
# Orders Paid by Boleto Are More Likely to Experience Late Delivery



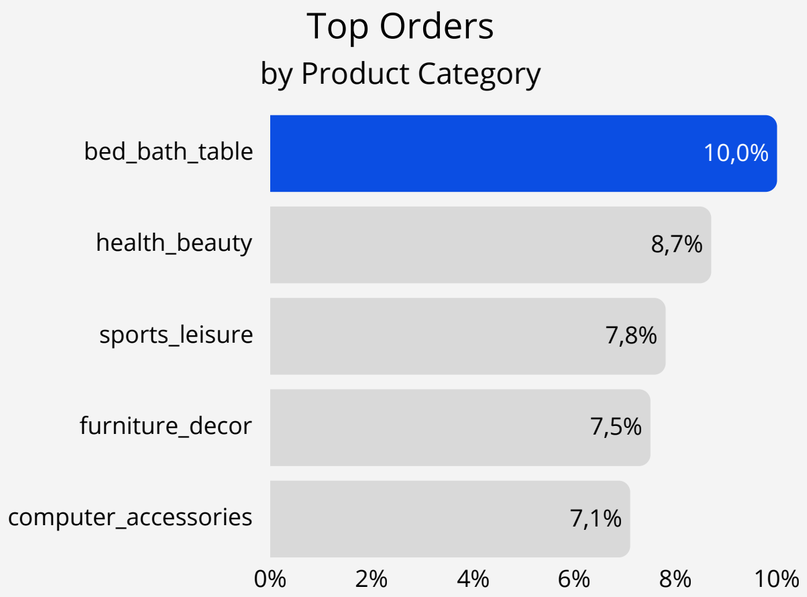

Payment method using boleto is only about 19% but has the highest late-delivery rate, suggesting that **longer payment approval times are associated with higher late-delivery risk.**

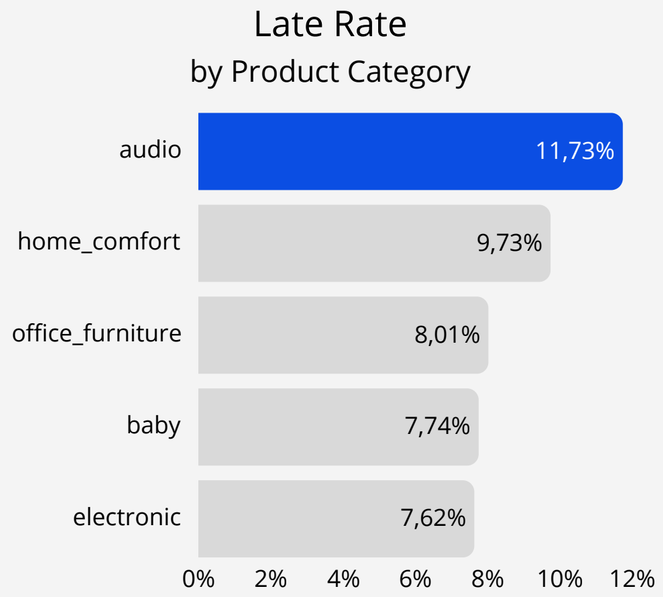
# Orders Paid with Longterm Are More Likely to Experience Late Delivery




- Credit card has the second-highest late-delivery rate (77.06% of total order) and **orders with long-term installments have the highest late-delivery rate** at 8.3%, compared with 6.7% overall
- Encourage customers to choose shorter installment periods, such as 1-month/full-payment options, to potentially reduce delivery risk.

# Late-Delivery Risk Varies Across Product Types and Characteristics




| CHARACTERISTIC | LATE | ON TIME | RESULT |
|---|---|---|---|
| Weight (median) | 800 g | 700 g | Significant |
| Length (median) | 25 cm | 25 cm | Significant |
| Height (median) | 13 cm | 13 cm | Significant |
| Width (median) | 20 cm | 20 cm | Not Significant |

- Order volume alone does not explain late-delivery risk.
- **Late-delivery rates vary across product categories**, with some categories with larger/heavier products showing higher risk.
- Statistically, **differences in product characteristics are significant** with late delivery although it's relatively small

# DATA SPLIT

**MAIN DATASET**
***95070 orders***

**TRAIN**

**60%** (57093 orders)

**Distribution of Target Variable (y_train)**




**VALIDATE**

**20%** (18867 orders)

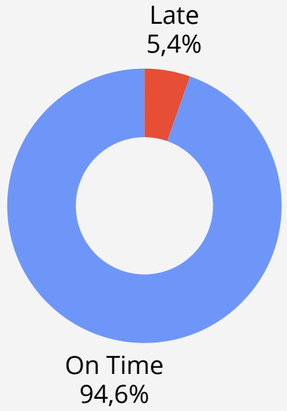
**Distribution of Target Variable (y_val)**




**TEST**

**20%** (19110 orders)

**Distribution of Target Variable (y_test)**




- dataset was splitted by using time to keep the integrity of the data being historical data

# FEATURE ENGINEERING

**Transform Raw Data Into Meaningful Signals for Delivery-Risk Prediction.**

**DERIVED FEATURES**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| item_volume_cm3 | integer | Combined product length × width × height |
| distance_group | category | Haversine distance between seller and customer zip-code centroids, also grouped into distance ranges |
| purchase_dow | string | Day of the week when the order was approved |
| purchase_month | string | Month when the order was approved |
| purchase_hour | integer | Hour when the order was approved |
| promised_days | integer | Number of days from order creation to estimated delivery date |
| is_black_friday_or_holiday | boolean | Indicates whether the order was placed during Black Friday or the December holiday season |
| is_interstate | boolean | Indicates whether the order crosses state boundaries between seller and customer |
| seller_late_rate | integer | Specific seller late to carrier rate |
| route | string | Specific seller-state to customer-state route |

**AGGREGATED ORDER**

| COLUMN | TYPE | DESCRIPTION |
|---|---|---|
| dominant_product_category | string | For multi-items, product with the highest number of items within the order |
| primary_seller_geo_state | string | For multi-items, we took the dominant seller’s state |
| primary_seller_late_rate | string | For multi-items where the sellers are different, we took the dominant |

# FEATURE TRANSFORMATION

**Transform features into a model-ready format**

### IMPUTATION

1. **Numerical: Median**
   'total_weight_g',
   'total_volume_cm3'

2. **Categorical: "missing"**
   ‘product_category‘

### SCALLING

**Robust Scaller**
applied to numerical features

### ENCODING

1. **One Hot Encoding**
   ‘purchase_dow’,
   ‘purchase_month’,
   ‘payment_type’,
   ‘distance_group’

2. **Binary Encoding**
   'route', '
   ‘seller_geo_state',
   'customer_geo_state',

### HANDLE IMBALANCE

**‘SMOTE’**
**‘SMOTEENN‘**
**‘SMOTETomek’**

# FEATURE SELECTION

**Select The Most Informative Features to Build A More Efficient Model**

**SelectKBest (f_classif)**

Selects the top 20 features based on ANOVA F-test scores
Applied before benchmarking linear-based models

**5-fold CV + Average Precision**

# METRIC EVALUATION

Objective:

**Identifies as many as possible late orders while minimazing too many false alarms**

| | FP | FN |
|---|---|---|
| Meaning | Predicted **Late** → Actually **On-time** | Predicted **On-time** → Actually **Late** |
| Business Impact | Unnecessary intervention | Missed late-delivery order |
| Estimated Cost ($R) | 0,04 | 42,5 |

Metric Used:

**PR-AUC - RECALL**

- the data is heavily imbalanced
- is not influenced by the large number of on-time orders (ignore true negative)
- focus on late orders (minority class)
- can controls unnecessary intervention (control false positive)

# MODEL BENCHMARKING

**Benchmarked 12 models using 5 Fold Time Series Cross-Validation; LightGBM achieved the highest Average Precision.**

**LINEAR BASED MODEL**

| MODEL | MEAN TRAIN SCORE | STD TRAIN SCORE | MEAN TEST SCORE | STD TEST SCORE | PRECISION | RECALL | F2 SCORE | PR-AUC | FIT TIME (S) |
|---|---|---|---|---|---|---|---|---|---|
| Linear SVM (LinearSVC) | 0,188 | 0,008 | 0,173 | 0,012 | 0,117 | 0,658 | 0,199 | 0,173 | 0,239 |
| Ridge Classifier | 0,188 | 0,007 | 0,173 | 0,012 | 0,117 | 0,657 | 0,198 | 0,173 | 0,218 |
| Bagged Logistic Regression | 0,186 | 0,008 | 0,172 | 0,012 | 0,118 | 0,658 | 0,200 | 0,172 | 0,783 |

and 2 more linear models records, including Logistic Regression and SGD Classifiier

**TREE BASED MODEL**

| MODEL | MEAN TRAIN SCORE | STD TRAIN SCORE | MEAN TEST SCORE | STD TEST SCORE | PRECISION | RECALL | F2 SCORE | PR-AUC | FIT TIME (S) |
|---|---|---|---|---|---|---|---|---|---|
| LightGBM | 0,492 | 0,148 | 0,240 | 0,015 | 0,495 | 0,097 | 0,492 | 0,757 | 1,016,908 |
| XGBoost | 0,765 | 0,141 | 0,223 | 0,014 | 0,446 | 0,099 | 0,765 | 0,742 | 0,876 |
| Random Forest | 1,000 | 0,000 | 0,209 | 0,014 | 0,417 | 0,102 | 1,000 | 0,733 | 6,827,506 |

and 4 more tree based models records, including GBM, Extra Trees, AdaBoost and Decision Tree

# HYPERPARAMETER TUNING

**Improve The Configuration of The Selected LightGBM Model**

| STEP | MODEL | METHOD | ITERATIONS | CV SPLITS | MEAN CV SCORE (AP) | VALIDATION PR-AUC |
|---|---|---|---|---|---|---|
| 1 | LightGBM | RandomizedSearch #1 | 50 | 5 | 0,239 | |
| 2 | LightGBM | RandomizedSearch #2 | 50 | 5 | 0,242 | |
| 3 | LightGBM | RandomizedSearch #3 | 50 | 5 | 0,239 | |
| 4 | LightGBM | GridSearch | 32 | 5 | 0,242 | 0,166 |

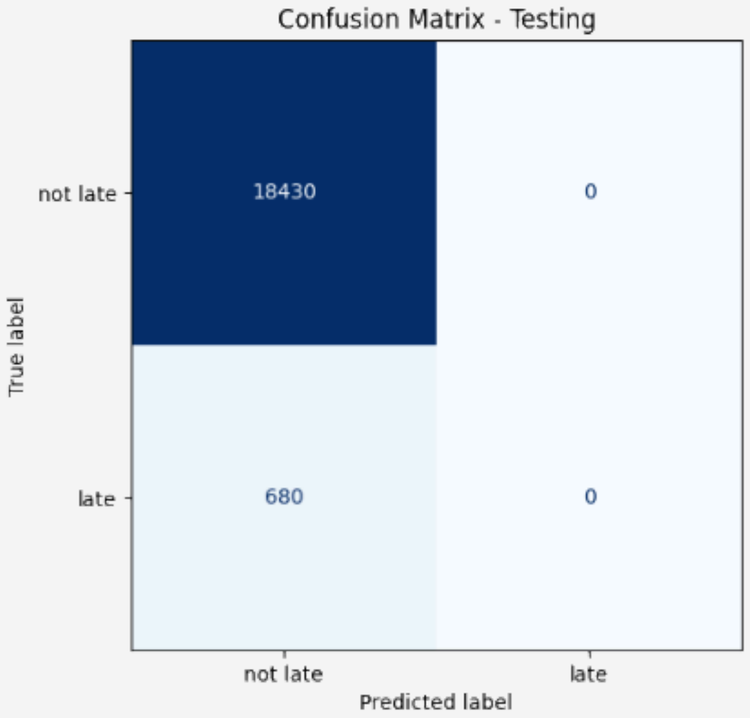
| PARAMETER | BEST VALUE |
|---|---|
| Sampling | SMOTE |
| n_estimators | 400 |
| num_leaves | 15 |
| max_depth | 4 |
| learning_rate | 1 |
| min_child_samples | 100 |
| subsample | 7 |
| colsample_bytree | 7 |
| reg_alpha | 0 |
| reg_lambda | 5 |

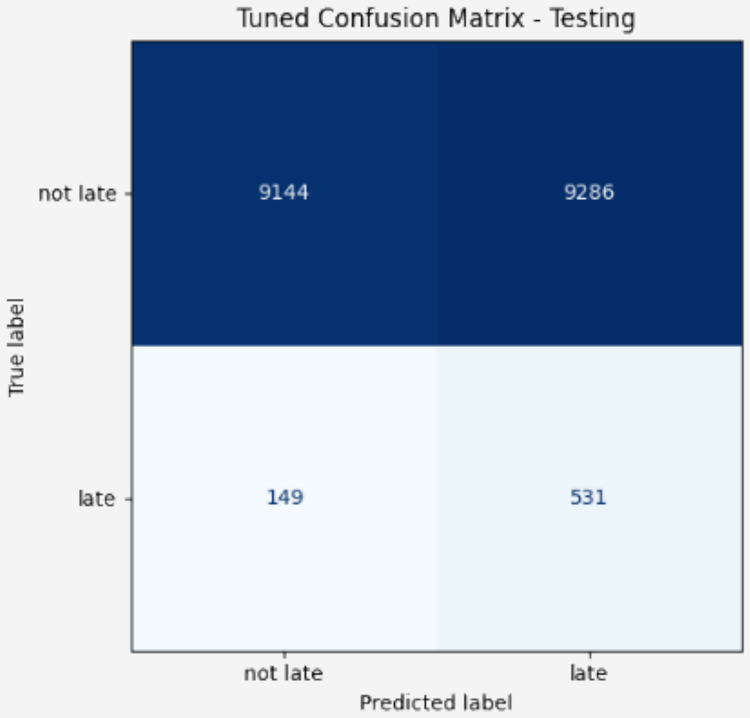
**Tuning Strategy**

- Random Search: find promising parameter ranges
- Grid Search: fine-tune the selected ranges
- 5-fold TimeSeriesSplit
- Metric: PR-AUC

# THRESHOLD OPTIMIZATION

**The Optimal Probability Threshold Achieves The Trade-off Between Recall and False Alarms**



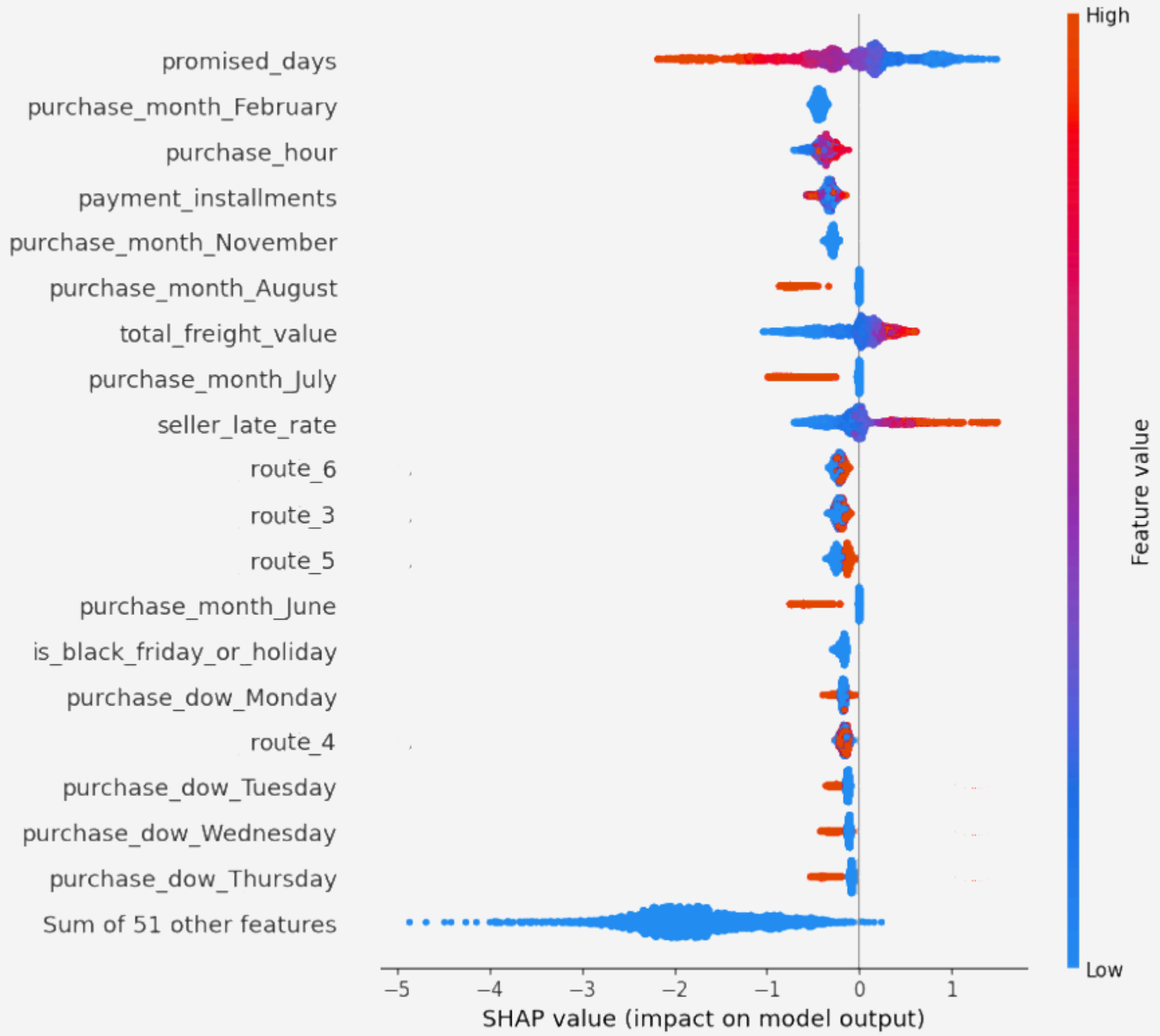




| Metrics | Default | Optimized (0.076) |
|---|---|---|
| Precision | 0%* | 13% |
| Recall | 0% | 87% |
| **F2-Score** | **0%** | **40%** |

*At the default threshold of 0.5, the model generated no positive predictions, resulting in 0% recall.

# FEATURE IMPORTANCE

**SHAP Analysis Reveals Several Key Features Driving High Late-Delivery Risk**




- **Promised Days:**

Shorter delivery promised days → higher late risk;
Longer promised days → lower late risk

- **Seller Late Rate:**

Higher seller late rates dispatch order to carrier → higher late risk
Lower rates → lower late risk.

- **Total Freight Value:**

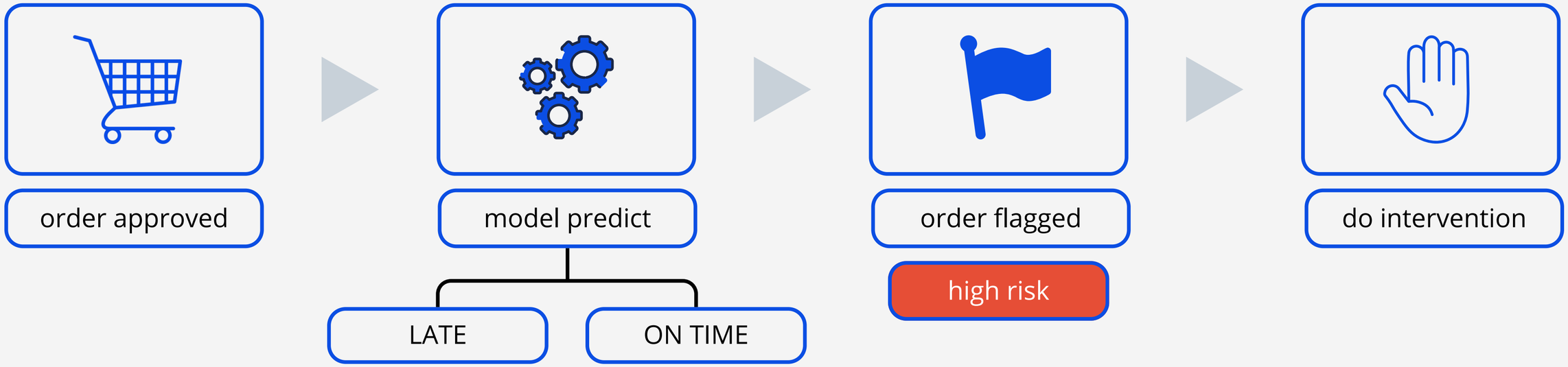
Higher freight value, which may reflect larger/heavier shipments, tends to increase late-delivery risk.

- **Purchase Month:**

February & November tend to increase late risk, while June–August tend to reduce it, suggesting potential seasonal/event effects.

# MODEL IMPLEMENTATION

**The Model Is Implemented to Identify and Flag High-risk Orders**



# MODEL LIMITATIONS

**The Model Still Has Limitations in Predicting Late Deliveries**

### DATA

Limited to Oct 2016–Aug 2018, with low early-month volume, limiting seasonal pattern capture.

### FEATURE

Lacks real-time/contextual signals and recent seller performance indicators.

### MODEL BEHAVIOR

Relies on 50+ features, making the prediction harder to translate into simple business rules.

### BUSINESS / DEPLOYMENT

- Requires retraining to adapt to new sellers, routes, or operational changes.
- Higher recall also increases false positives and unnecessary interventions.

# The Model Delivers Positive Profit While Avoiding The High Seller-Fatigue Cost of Notifying Everyone

| Strategy | Notification Cost ($R) | Success Profit ($R) | Failure Loss ($R) | Net Value ($R) | Ignore Cost ($R) |
|---|---|---|---|---|---|
| Do Nothing (Baseline) | 0 | 0 | 28.900 | -28.900,00 | 0 |
| Using Model | 441,76 | 22.567,5 | 6.332,5 | 15.793,24 | 131.582,62 |
| Notify Everyone | 859,95 | 28.900 | 0 | 28.040,05 | 261.153,10 |

- **Using the predictive model**: shifts the outcome from a net loss of R$ -28,900 to a net profit of R$ 15,793.24, showing that targeted intervention can **create positive business value**.
- **Notify everyone**: generates the highest net value, but also creates the highest ignore cost from seller fatigue due to unnecessary interventions.
- **Using the model** provides the **better balance** between **net value and ignore cost**, by focusing intervention on orders with higher late-delivery risk rather than notifying all sellers.

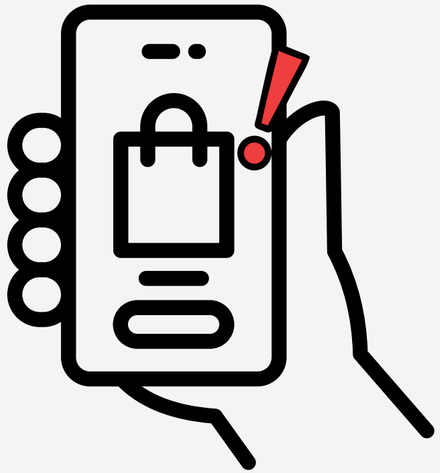
# The Model Helps Improve Customer Satisfaction by 32.26% with A Better Business Trade-Off

| Strategy | Expected AVG Review Score | Rating Increase | Total Increase (%) |
|---|---|---|---|
| Do Nothing (Baseline) | 3,01 | 0 | 0 |
| Using Model | 3,98 | 0,97 | 32,26 |
| Notify Everyone | 4,00 | 0,99 | 32,78 |

- **Using model**: Review score increases from 3.01 to 3.98, representing a 32.26% improvement.
- **Notify everyone**: Achieves a slightly higher 32.78% improvement, but the additional gain is only 0.52% points compared with the model.
- **Business trade-off**: After considering the intervention cost, the model remains the better option, delivering nearly the same improvement without intervening on every order.

# CONCLUSION

## Factors Associated with High Risk Orders:



- Short promised days
- Seller with higher late rate to carrier
- Big total freight value
- Purchased in seasonal events

## How Latency Dictates Customer Ratings:



customers ratings drop significantly for late orders but remain stable as long as the promised delivery window is met

By using the model, Olist could:

- 🔻 **Reduce late delivery** from 5.4% → 1.18%
- 🔺 **Improve average review score** by 32,26% from 3.01 → 3.98
- 🔺 **Generate net value** R$15,793.24

# MODEL RECOMMENDATION

**Strengthen the model with external data that captures seasonal, environmental, geographic, and logistics-related delivery risks.**

### Traffic & Seasonal Disruption Data

Add seasonal calendar and road-closure data to capture predictable disruptions such as Carnival-related traffic and road closures.

### Real-Time Weather Forecast Data

Integrate weather forecasts available at the time of purchase to capture potential disruptions from storms, flooding, and other severe weather

### Infrastructure & Terrain Data

Add road quality, elevation, and urban/rural classification to better capture route accessibility beyond delivery distance.

### Logistics Partner & Carrier Metrics

Incorporate carrier performance, vehicle capacity, and operational metrics through data-sharing with logistics partners or sellers.

# BUSINESS RECOMMENDATION

**Proactive Intervention Workflow**

**flag high risk orders**



Use the predictive model to identify orders with a high probability of late delivery.

**notify sellers**



Send an automated notification to the seller requesting that they adhere to the shipping buffer days to prevent further delays.

**track shipment**



Monitor flagged shipments and coordinate with logistics partners when an order is at risk of exceeding the promised delivery date.

**adjust estimated delivery day**



Adjusting the estimated delivery date based on delivery distance

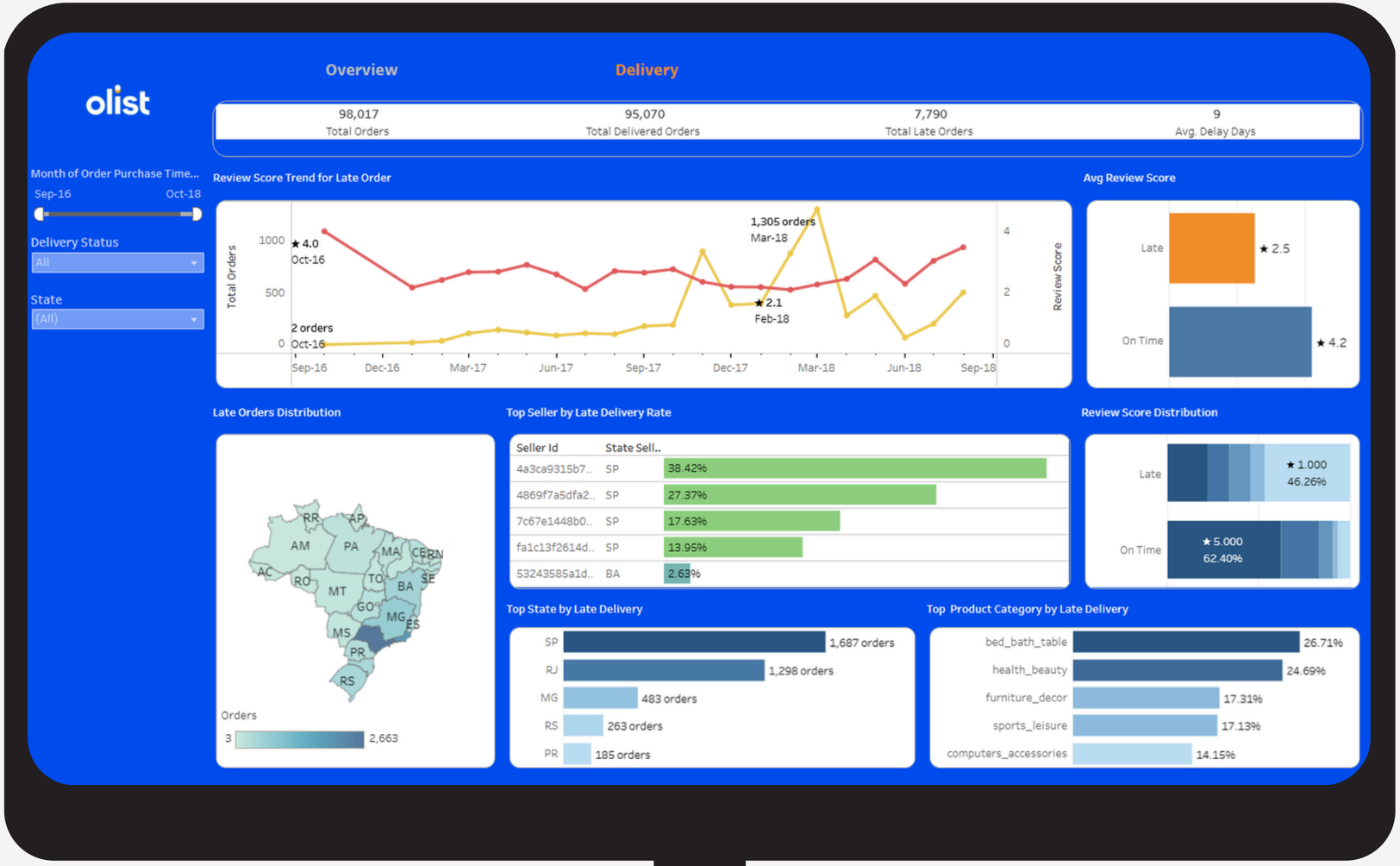
- short +4 days
- Regional +5 days
- interegional +7 days
- Long haul +8 days

**notify customers**

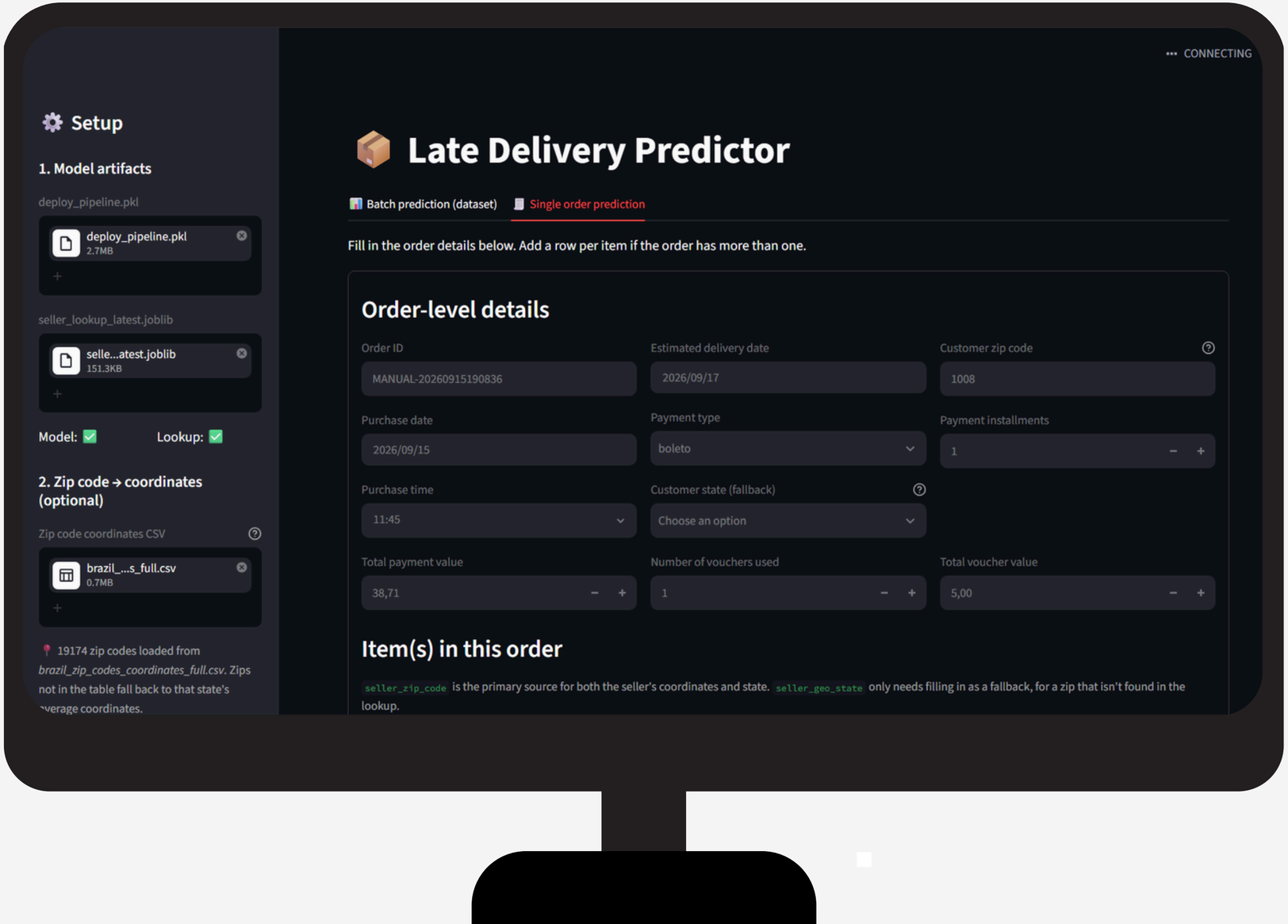


Proactively inform affected customers about potential delays and provide a channel to address their questions or concerns.

# TABLEAU DASHBOARD



# STREAMLIT APP DEPLOYMENT



# THANK YOU!



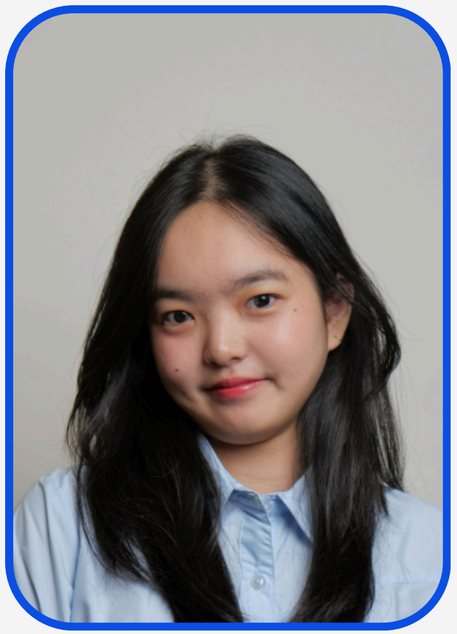
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